

MPTE: Theory-Validation Simulation

Results report (internal, for discussion)

July 1, 2026

1 What this validates

The inferential theory (Theorems 1–3) is derived for the *linear* MPTE estimator: PCA on the attention-weighted panel $\tilde{Z} = BZA_z$ with fixed operators B , A_z . This report validates that estimator on a controlled union-factor linear DGP with known ground truth, and bridges it to the nonlinear model. Four experiments: E1 consistency (Theorem 1), E2 coverage (Theorems 2–3), E3 efficiency from transfer learning, E4 nonlinear bridge.

Model note: recovering the two attention operators. The theory uses two operators, a temporal B ($T \times T$) that reweights time and a cross-sectional A_z ($N \times N$) that reweights variables. A single self-attention layer produces only *one* square matrix, over whatever we treat as its tokens: attend over the T time rows and we get a $T \times T$ matrix (B); attend over the N variable columns and we get an $N \times N$ matrix (A_z). One attention therefore sees one axis only. The empirical pipeline flattens the whole panel into one long token stream, runs one large attention over it, and recovers B and A_z afterward by averaging that attention over variables and over time. At the sizes needed to trace asymptotic rates the flattening is infeasible: a flattened $T \times N$ panel has $T \cdot N$ tokens and a $(T \cdot N)^2$ attention, for example 160,000 tokens and 2.56×10^{10} entries at $N = T = 400$. We therefore use *axial* attention: one attention over the N columns yields A_z and one attention over the T rows yields B , directly and at cost $N^2 + T^2 \approx 3.2 \times 10^5$. This produces the same two operators the paper defines, obtained directly rather than by averaging a flattened map. The value map is fixed to the identity ($W^v = I$, so $V = Z$): the attention reweights the panel Z itself, which keeps $\tilde{Z} = BZA_z$ an approximate factor model in the same factors and loadings, so PCA on $\tilde{Z}$ recovers them.

Setup. The data-generating process is the union-factor linear model of the design note ($k = 4$ factors, $N_x:N_y = 2:1$, iid idiosyncratic noise). Every quantity averages over 2000 Monte Carlo replications: per (T, N) cell in E1 (grid $N = T \in \{80, 160, 320, 640, 1000\}$), in E2 and E3, and over 2000 out-of-sample panels in E4 (at $N = T = 300$).

2 E1 — Consistency (Theorem 1)

Theorem 1 gives consistency of the factor, loading, and common-component estimates. We track the common component $C = F\Lambda^\top$, the rotation-free combination of the factor and loading estimates: it needs no rotation alignment and summarizes both in one quantity. We report its relative error $\|\hat{C} - C\|_F^2 / \|C\|_F^2$ as the panel grows along $N = T$. The relative form is scale-invariant, so it is comparable across operator arms, whereas the absolute error confounds estimation quality with the operator-dependent scale of the target $C = BCA_z$. Consistency requires the relative error to decline toward zero, and a log-log slope near -1 matches the classical PCA rate $1/T + 2/N$.

We report three operator arms, each grounded in the paper. The oracle arm fixes $A_z = I$ and $B = I$, the pure-theory reference. The parameter-free arm uses the paper’s simplest attention, $Q = K = Z$, so that $A_z = \text{softmax}(Z^\top Z/\cdot)$ carries no learned projection. The learned arm reads the frozen, scaled attention off the trained linear model, estimated on a separate panel and then held fixed, the sample-splitting convention of the paper’s Remark 1. All three arms converge at a slope near one (Figure 1, Table 1), so they agree, which is the property the design asks for: the data-driven operators inherit Theorem 1 at the same rate as the oracle.

A note on Assumption A.7. For the two data-driven arms the cross-sectional operator norm $\|A_z\|_{\text{op}}$ grows with N rather than staying bounded, so A.7 as stated does not hold along this path. The convergence rate is nevertheless unaffected, as panel (d) read against panels (a)–(c) shows. We read this as evidence that A.7 is sufficient but stronger than necessary in these designs. Whether it can be relaxed to a slowly growing operator norm, with the rate re-derived, is a question for the theory that we leave open here.

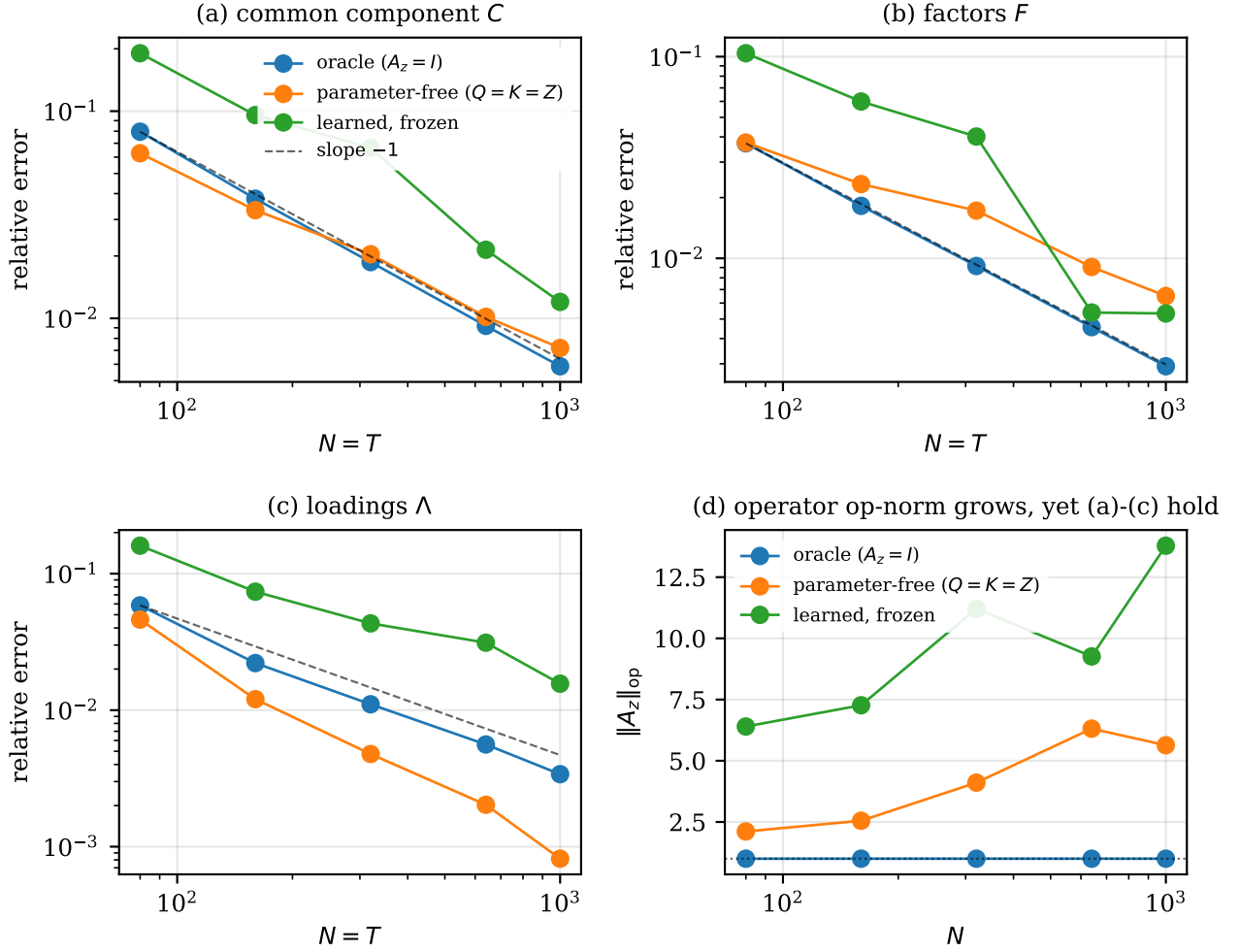


Figure 1: E1: relative estimation error against $N = T$ on log-log axes for (a) the common component C , (b) the factors F , and (c) the loadings Λ , each with a slope near -1 for all three operator arms; (d) the cross-sectional operator norm $\|A_z\|_{\text{op}}$, which grows for the data-driven arms while the rates in (a)–(c) hold.

Operator arm	slope C	slope F	slope Λ	$\ A_z\ _{\text{op}}$ at N_{max}
oracle ($A_z=I$)	-1.03	-1.01	-1.10	1.0
parameter-free ($Q=K=Z$)	-0.86	-0.69	-1.53	5.6
learned, frozen	-1.08	-1.30	-0.85	13.8

Table 1: E1: fitted slope of the relative error against N (target -1) for the common component, factors, and loadings, per operator arm, with the operator norm at the largest N .

3 E2 — Coverage / asymptotic normality (Theorems 2–3)

Empirical coverage of CIs for the Y -strong common component $C_{y,i,t}$, using the analytic plug-in variance from the paper. For the baseline iid DGP the long-run covariances collapse to $\Omega = \sigma^2 \Sigma_{F,y}$, $\Xi = \sigma^2 \Sigma_{\Lambda,y_s}$, and we studentize with the general Theorem-3(iii) variance $\text{Var}(\hat{C} - C) = \sigma_F^2/N + \sigma_\Lambda^2/T$ (the F - and Λ -dominant cases are its limits). Coverage is near nominal in all three regimes (Table 2) and the studentized statistic is standard normal (Figure 2).

Regime	N	T	Cov. 90%	Cov. 95%	z sd
F-dominant	50	400	0.878	0.936	1.057
Lambda-dominant	400	50	0.886	0.941	1.021
mixed	200	200	0.894	0.947	1.006

Table 2: E2: empirical coverage of 90%/95% CIs and the studentized-statistic sd, by regime.

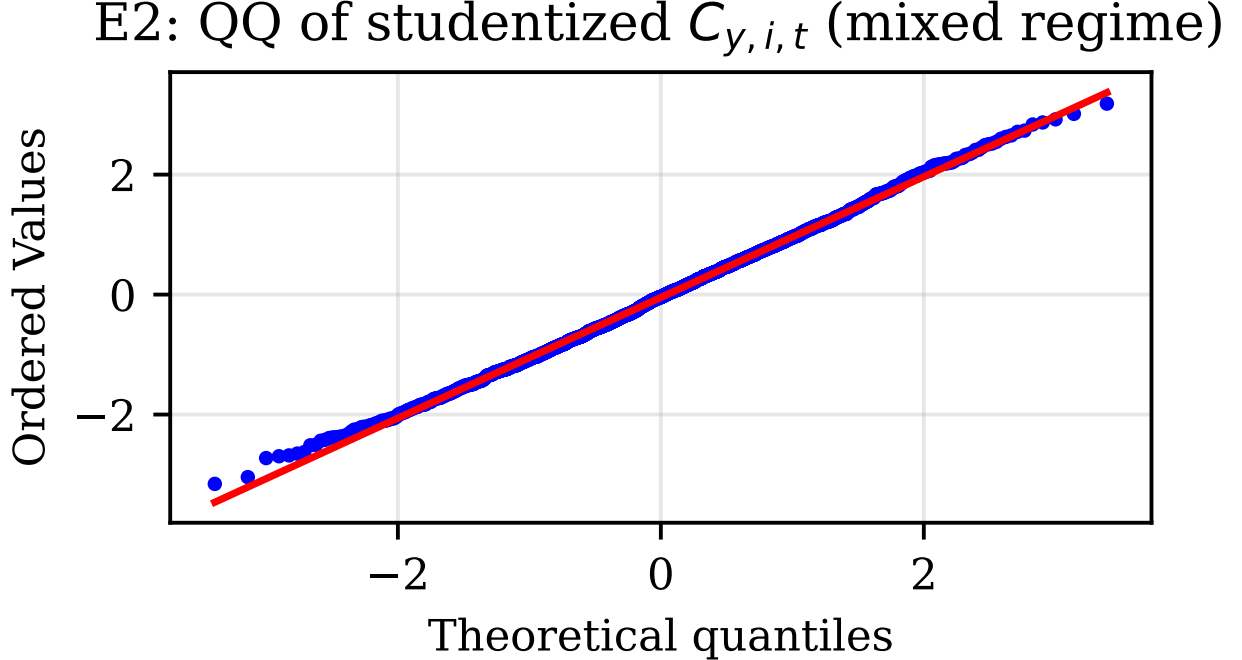


Figure 2: E2: QQ plot of the studentized $C_{y,i,t}$ against $\mathcal{N}(0, 1)$ (mixed regime).

4 E3 — Efficiency from transfer learning

The joint (X, Y) estimator of $C_{y,i,t}$ is more precise than the Y -only estimator; the gain grows with the auxiliary size N_x and saturates because only one factor is shared (Figure 3, Table 3).

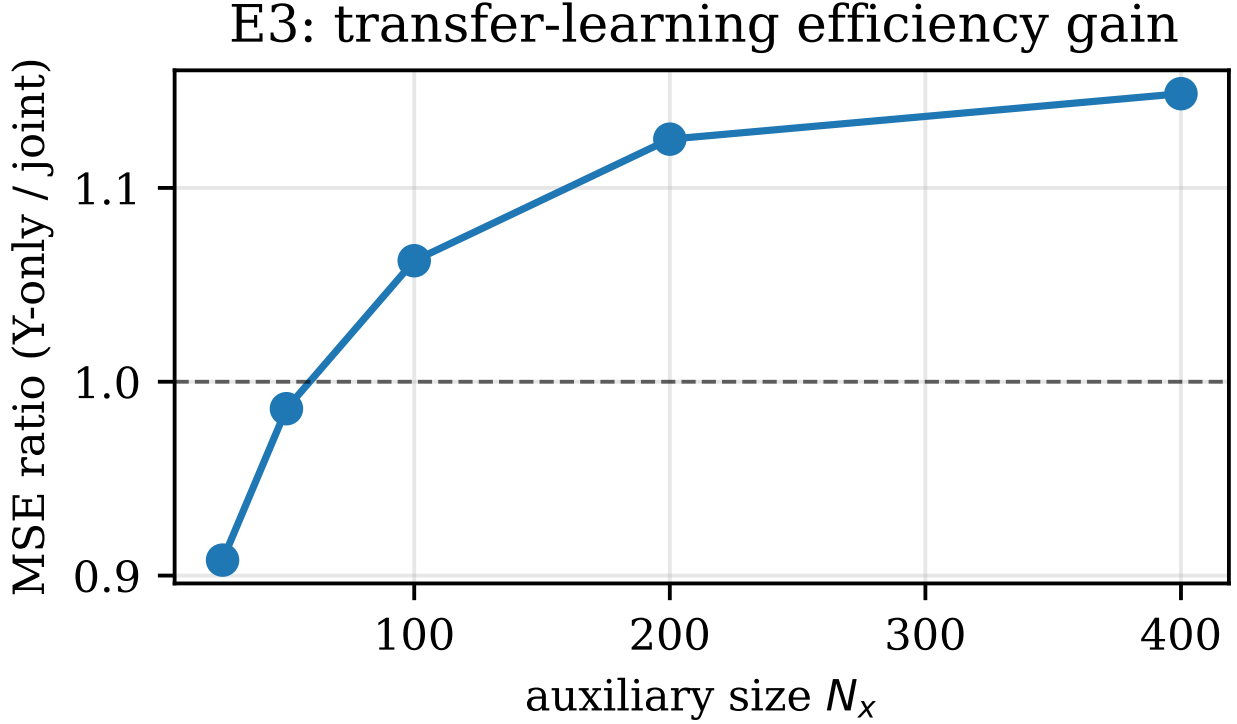


Figure 3: E3: MSE ratio (Y-only / joint) vs. N_x .

N_x	MSE joint	MSE Y-only	ratio
25	0.1119	0.1016	0.908
50	0.1032	0.1018	0.986
100	0.0955	0.1015	1.062
200	0.0905	0.1018	1.125
400	0.0884	0.1015	1.149

Table 3: E3: joint vs. Y-only common-component MSE and their ratio.

5 E4 — Nonlinear bridge

On linear-truth data we ask whether the nonlinear model builds the factor representation the theory describes (Figure 4). We measure two subspaces against the true attended factors $F^{(B)} = BF$ by canonical correlation: the nonlinear model’s k -dim latent, and the linear PCA estimator’s factors. The linear estimator recovers the full attended factor space. The nonlinear model, trained on a single-target forecast objective with a deliberately minimal architecture, recovers the target-relevant direction strongly (leading canonical correlation ≈ 0.98) while leaving the orthogonal factors to the linear estimator, since a single-target objective only needs the direction that predicts its target. The forecast-accuracy equivalence between the nonlinear model and the linear ablation is the paper’s

Table 1 result; here we add the factor-space statement against known ground truth for the target signal.

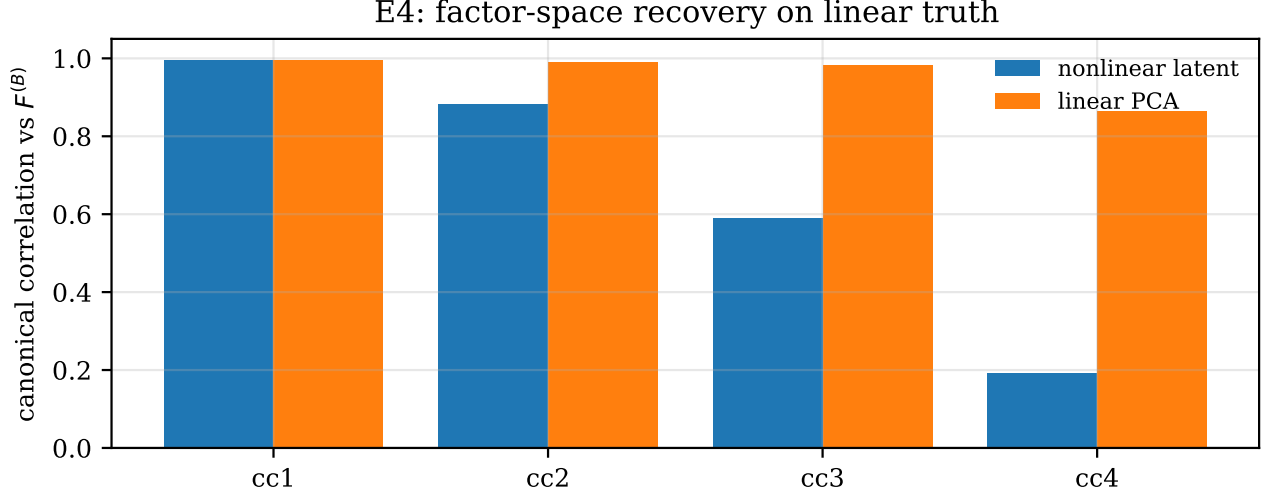


Figure 4: E4: canonical correlations with $F^{(B)}$, nonlinear latent vs. linear PCA. The nonlinear model recovers the leading, target-relevant factor; the linear estimator recovers the full space.

6 Learned operators

Figure 5 shows the frozen cross-sectional operator A_z and temporal operator B read off the trained linear model on one panel. The attention concentrates on a few variables and time points, the bright stripes, which is the structure behind the operator norm that grows with N in Figure 1(d).

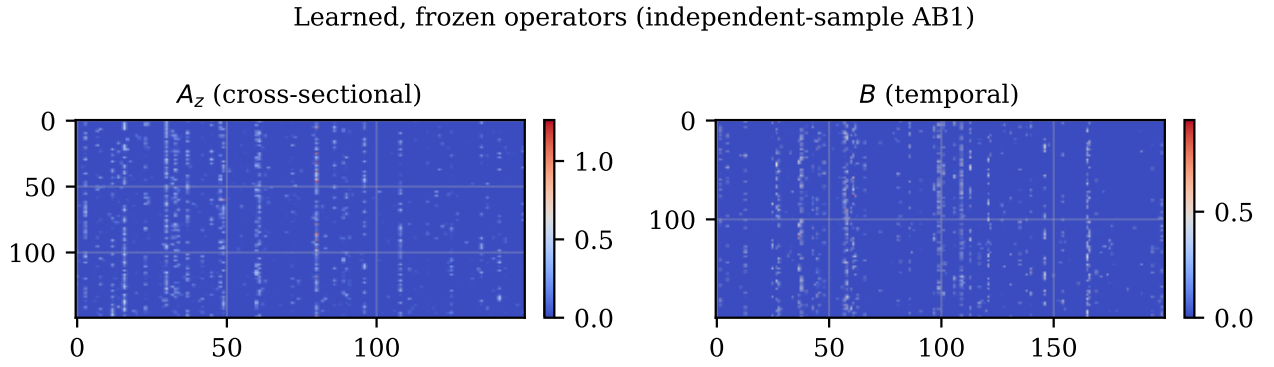


Figure 5: Learned, frozen A_z ($N \times N$) and B ($T \times T$) from the independent-sample linear model.